CHAPTER 1

Sequences and mathematical induction

S2, S4, S5, S7, S10, S11, S13

Before you start · Tracks and handwriting indicated

Without a calculator, in six minutes. Answer briefly. Mobilize the "reference sequences" sheet and E1·E2. Then choose the targeted review indicated under the questions.

1. Calculate u_1 then u_2 if $u_0 = 3$ and $u_{n+1} = 2u_n - 1$.
2. Distinguish u_{n+1} and $u_n + 1$.
3. Give $1 + 2 + \cdots + 10$.
4. Write $\sum_{k=0}^{3} 2^k$.
5. Determine the multiplier coefficient by a 20 per cent decrease.
6. Give the first three values of `n` in `for n in range(3)` .

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Replace successively the previous value; R01. **2.** Write an index row and an addition on the line; R01. **3.** Match first and last term; R09 to compare two sums. **4.** Developing the four terms; R08 control their number. **5.** Write 100 minus 20 out of 100; R03. **6.** List values of place before calculation; R07. R questions are used as support for resuming with the teacher, even if their collective ritual is planned later.

Objectives

Calculate terms and sum; draft an induction; establish monotonicity and bounds; operate an auxiliary sequence.

Course notes

Set a sequence and calculate

A real sequence is a function defined on $\mathbb{N}$, or on integers from a row n_0. We note u_n his term of rank n. An explicit formula gives u_n direct; a recurrence relationship requires also an initial term. $u_0 = 2$ and $u_{n+1} = 3u_n - 1$ give $u_1 = 5$, $u_2 = 14$.

Reference sequences

If $u_{n+1} = u_n + r$, then $u_n = u_p + (n-p)r$. If $u_{n+1} = qu_n$, then $u_n = u_p q^{n-p}$ for $n \geq p$.

$$\sum_{k=p}^{n} u_k = (n-p+1)\frac{u_p + u_n}{2} \quad \text{(arithmetic)}, \qquad \sum_{k=p}^{n} u_k = u_p \frac{1-q^{n-p+1}}{1-q} \quad (q \neq 1).$$

When $q = 1$, the second sum is $(n-p+1)u_p$.

Induction reasoning

To prove $P(n)$ for the integer $n \geq n_0$: check $P(n_0)$; fix an arbitrary integer $n \geq n_0$ and assuming $P(n)$ true; deduce from it $P(n+1)$; the conclusion by the principle of recurrence. not the conclusion: it can only be used at the fixed rank.

Completely written example. Let $u_0 = 1$, $u_{n+1} = 2u_n + 3$. For $n \in \mathbb{N}$,note $P(n)$: « $u_n = 4 \times 2^n - 3$ ».

Initialization. $4 \times 2^0 - 3 = 1 = u_0$; $P(0)$ is true.

Heredity. Fixes $n \in \mathbb{N}$ and suppose $P(n)$ True. By the definition of the future, then by this hypothesis,

$$\begin{aligned} u_{n+1} &= 2u_n + 3 \\ &= 2(4 \times 2^n - 3) + 3 \\ &= 8 \times 2^n - 3 = 4 \times 2^{n+1} - 3. \end{aligned}$$

It's $P(n+1)$: the property is hereditary.

Conclusion. By the principle of recurrence, $u_n = 4 \times 2^n - 3$ for all $n \in \mathbb{N}$.

Demonstration: sum of integers. Note: $P(n)$: « $\sum_{k=0}^{n} k = \frac{n(n+1)}{2}$ For $n = 0$, both members are 0: $P(0)$ It's true. $n \geq 0$ and suppose $P(n)$ True, then.

$$\begin{aligned} \sum_{k=0}^{n+1} k &= \sum_{k=0}^{n} k + (n+1) \\ &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{(n+1)(n+2)}{2}. \end{aligned}$$

We've established $P(n+1)$. By induction, the formula is true for everything $n \in \mathbb{N}$.

Overall behaviour

The rest is increasing if $u_{n+1} \geq u_n$ for any rank, decreasing if inequality is reversed. increased if there is a real M such that $u_n \leq M$ for every n ; minor if it exists m such that $u_n \geq m$; limitate if it is both reduced and increased. To study monotonicity, one can study the sign of $u_{n+1} - u_n$. If all the terms are strictly positive, we can compare $\frac{u_{n+1}}{u_n}$ 1. The study of a function f such that $u_n = f(n)$ is another possibility.

Arithmetico-geometric sequence

For $u_{n+1} = au_n + b$, $a \neq 1$, solve $\ell = a\ell + b$, then pose $v_n = u_n - \ell$. We get $v_{n+1} = av_n$, so

$$u_n = \ell + (u_0 - \ell)a^n, \qquad \ell = \frac{b}{1-a}.$$

If $a = 1$, the sequence is arithmetic: do not divide by 1 $- a$.

Example. If $u_0 = 10$ and $u_{n+1} = 0{,}8u_n + 6$, then $\ell = 30$ and $u_n = 30 - 20(0{,}8)^n$. We have $10 \leq u_n < 30$ and $u_{n+1} - u_n = 4(0{,}8)^n > 0$.

Trap. Checking a property on a hundred terms doesn't show it. A increasing sequence isn't necessarily positive; a bounded sequence is not necessarily monotonous.

Understand, try, resume

1.1 Rank, Value and Update

First use: S2 and S4.

Definition · A sequence is not a finished list

The index n means an integer; u_n is a real. The point of representation is $(n; u_n)$. One line connecting points can help reading but does not define terms to non-integer indices.

Solved Example · Two Similar Expressions

For $u_0 = 2$ and $u_{n+1} = 3u_n - 1$, we get $u_1 = 5$, $u_2 = 14$, $u_3 = 41$. Thus $u_2 + 1 = 15$ while that $u_{2+1} = 41$. In a loop, the variable `u` contains first u_n ; education `u = 3*u-1` replace it by u_{n+1}.

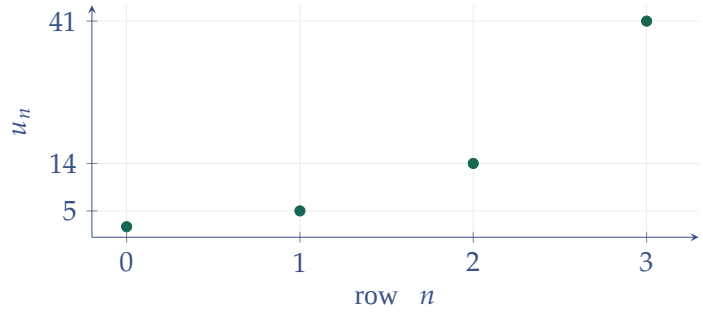


Figure 1.1. Read the term rank 2: his order is 14, not 2.

A1 · Immediate application For $v_0 = 1$, $v_{n+1} = 2v_n + 1$, build a table of rows 0 to 3, then place the four points. Calculate separately $v_2 + 1$ and v_{2+1}.

A2 · Partial guidance A loop must be calculated $u_0 + u_1 + u_2 + u_3$ for the rest of the example. of u=2; total=2 . How many updates do you need? Write the order of instructions and explain why adding before the update would count twice u_0.

1.2 A local hypothesis, a universal conclusion

First use: S5 and S7.

Example resolved · Recurrence of inequality

Let $w_0 = 1$ and $w_{n+1} = \frac{1}{2}w_n + 2$. Prove $P(n) : 1 \leq w_n < 4$. Initially $w_0 = 1 \in [1; 4[$. Fixes $n \in \mathbb{N}$ and suppose $P(n)$. Since $\frac{1}{2} > 0$, $\frac{1}{2} \leq \frac{1}{2}w_n < 2$, so $\frac{5}{2} \leq w_{n+1} < 4$. Like $\frac{5}{2} \geq 1$, this implies $P(n+1)$. The principle of recurrence provides guidance at all levels.

A3 · Proof to be completed For $a_0 = 2$, $a_{n+1} = 2a_n + 1$, complete: " $P(n)$ means $a_n = 3 \times 2^n - 1$. At range 0, ... Fixes . . . and suppose . . . . Then $a_{n+1} = \ldots = 3 \times 2^{n+1} - 1$. Therefore . . . "

A4 · Repair Evidence A student writes: "If $u_n \geq 2$, then $u_{n+1} = 2u_n \geq 4 \geq 2$. So all the terms are at least 2.' He did not give u_0. Build a counter-example, then indicate which given makes his proof complete.

1.3 Count before summation

First use: S10.

Property · Number of terms

For integers $p \leq n$, the sum $\sum_{k=p}^{n} u_k$ contains $n - p + 1$ terms. It can be verified in Writing first and last, before any formula.

Example resolved · A offset geometric sum

In $\sum_{k=3}^{7} 2^k$, there are five terms and the first is worth 8. The sum is 8 $\frac{1-2^5}{1-2} = 248$. For Demonstrate the formula without memorizing it in isolation, write $S = 1 + q + \cdots + q^n$, then subtract $qS = q + \cdots + q^{n+1}$: $(1-q)S = 1 - q^{n+1}$. The division presupposes $q \neq 1$.

A5 · Independent evidence Demonstrate by induction, for everything $n \in \mathbb{N}$, $\sum_{k=0}^{n}(2k+1) = (n+1)^2$. Explain why the term added in heredity is 2 $n + 3$, and not $2n + 1$.

1.4 Recognize a structure without confounding it with a boundary

First use: S11 and S13.

Example solved · Choose the offset

For $w_{n+1} = \frac{1}{2}w_n + 2$, the fixed value is 4. By subtracting 4 of the two members, $w_{n+1} - 4 = \frac{1}{2}(w_n - 4)$. Thus $w_n = 4 - 3\,2^{-n}$. The gap is not arbitrary: it makes the term disappear This explicit calculation does not yet imply a convergence theorem.

A6 · Choice of strategy A subscription costs 40 euros in the first year, then its cost increases each year by 5 euros. Another costs 40 euros and then increases by 10 % per year. Write both to compare the first three years and calculate the cumulative costs over ten years. number of terms.

A first step: First name the nature of the two developments. The tenth year corresponds to the rank 9 if the first is row 0.

Wait a minute at the courthouse. Explain the difference between "for everything n, $u_n \leq 4$" and "there exists n such that $u_n \leq 4$". Give a follow-up that checks the second but not the first.

Review postponed · S20 and then S41 Take over $w_n = 4 - 3\,2^{-n}$ to prove its convergence; in probability, look again for the shift when a probability checks a relationship affine.

Research activity

One tank powered every day · 35 min.

A tank initially contains 100 litres of water. Every day, 20% of the present volume is taken, Then we add 30 liters. For all natural n, we note u_n the volume of water, in litres, after these two transactions n So, $u_0 = 100$.

1. Calculate u_1 and u_2, then express u_{n+1} according to the u_n.

2. What value of the volume would remain unchanged after a day? Demonstrate by induction that, for every $n \in \mathbb{N}$, $100 \leq u_n < 150$.

3. For every $n \in \mathbb{N}$, we're laying $v_n = u_n - 150$. Show that (v_n) is geometrical. terms of v_n and u_n according to the n.

4. Write a Python function without argument that returns the first integer n for which $u_n > 145$. Run the program and interpret the result.

5. One student calculated the first hundred terms and found that they were less than 150. Is it enough to show that all the terms are? Justify.

Progressive exercises

E1 · Direct application. What's next? (u_n) defined, for all natural $n$by $u_n = 5 - 3n$. Calculate u_4, then the sum $\sum_{k=0}^{20} u_k$.

E2 · Application and justification. Next (v_n) is defined by $v_0 = 3$ and, for the integer natural n, $v_{n+1} = 2v_n$. Express v_n according to the n, then calculate $\sum_{k=0}^{n} v_k$.

E3 · Method to be organised. Demonstrate by induction that, for the integer natural n, $3^n \geq 2n + 1$.

E4 · Choice of strategy. What's next? (u_n) defined on $\mathbb{N}$ by $u_n = \frac{n+1}{n+2}$. Demonstrate that it is increasing and giving a framework valid for all its terms.

E5 · Synthesis or demonstration. Demonstrate by induction that, for the integer natural n, $8^n - 1$ is divisible by 7. We will not use the congruences.

Work in the manual or annal

Barbazo, p. 28, Exercises 22 and 25. In class, in S5: exercise 22, to translate properties of sequences with a quantifier. At home after S11: exercise 25 integer, to articulate recurrence, add up and monotonicity. Write separately initialization, inheritance and conclusion.

Duty at home

DM1 · One sum and one threshold · 45 min, after S11.

What's next? (u_n) defined by $u_0 = 2$ and, for everything $n \in \mathbb{N}$, $u_{n+1} = \frac{1}{2}u_n + 3$.

1. Demonstrate by induction that 2 $\leq u_n < 6$ for all $n \in \mathbb{N}$. Deducing the direction of variation of (u_n).

2. Set $v_n = u_n - 6$ for all $n \in \mathbb{N}$. Show that (v_n) is geometric, then express u_n in function of n.

3. For all $n \in \mathbb{N}$, we note $S_n = \sum_{k=0}^{n} u_k$. Express S_n according to the n.

4. Using the explicit formula and powers of 2, determine the smallest integer n such that $u_n > 5.99$. Check that the integer previous one is not suitable.

5. Express 6 $- \frac{S_n}{n+1}$ and justify that this difference is strictly positive. Interpret the result in Whereas the average $n + 1$ first terms; no limit calculation is required.

MCQ A · consolidation (after S7)

After S7: first recurrences.

For each question, only one proposal is accurate.
consult the correction.

1. If $u_0 = 2$ and $u_{n+1} = u_n + 3$, then u_2 is valid for:

A. 5 B. 8

C. 11 D. 2^4

2. To demonstrate $P(n)$ by induction from 0 onwards, the inheritance consists of:

A. Supposing $P(n+1)$ B. Check only $P(0)$

C. Supposing $P(n)$ then prove $P(n+1)$ D. Solve $P(n) = 0$

3. The sum $\sum_{k=0}^{4} 2^k$ includes:

A. 4 terms B. 5 terms

C. 8 terms D. 2^4 terms

4. If $u_{n+1} - u_n > 0$ for all n, then:

A. u is increased B. u is increasing

C. u converge D. $u_n > 0$

5. In `while u <= 10: u = u + 1`, the buckle stops when:

A. $u = 10$ B. $u < 10$

C. $u > 10$ D. $u < 0$

MCQ B · Review and Method Choice (after S20)

After S20: monotonous convergence and auxiliary sequence.

For each question, only one proposal is accurate.
consult the correction.

6. A continuation is increasing and increased by 7. It can be concluded that:

A. tends to 7 B. converge

C. is positive D. is geometric

7. For $u_{n+1} = 0{,}5u_n + 4$, the potential equilibrium value checks:

A. $L = 0{,}5L + 4$ B. $L = u_0$

C. $L = 0{,}5 + 4L$ D. $L = 4^n$

8. If $v_n = u_n - 8$ and $u_{n+1} = 0{,}5u_n + 4$, then:

A. $v_{n+1} = v_n + 4$ B. $v_{n+1} = 0{,}5v_n$

C. $v_{n+1} = 8v_n$ D. $v_{n+1} = u_n$

9. « u_n belongs to $]5{,}99; 6{,}01[$ " is equal to:

A. $u_n < 6$ B. $|u_n - 6| < 0{,}01$

C. $|u_n| < 6$ D. $u_n > 0{,}01$

10. A simulation of 20 terms proves that:

A. The property is true for everything n B. the sequence converges

C. the 20 calculated terms have the observed property D. recurrence is useless

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. Any increasing sequence converges.

2. If $u_n = 6 - 2^{-n}$, then $u_n < 6$ for all n.

3. Solve $L = f(L)$ is sufficient to demonstrate that any follow-up $u_{n+1} = f(u_n)$ converge.

Review review

I can calculate a term, a sum, build an induction and study a monotonicity by $u_{n+1} - u_n$.
To be monitored: confuse a rank and a value; turn a digital observation into evidence.